

Air System Sizing Summary for HP-101-FITNESS

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-101-FITNESS**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **1279.5** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **3.2** Tons
Total coil load **38.9** MBH
Sensible coil load **34.5** MBH
Coil CFM at peak load **1215** CFM
Sum of peak zone CFM **1215** CFM
Sensible heat ratio **0.887**
CFM/Ton **374.7**
sqft/Ton **394.6**
BTU/(hr sqft) **30.4**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **82.0 / 64.5** F
Leaving DB / WB **53.4 / 53.3** F
Resulting RH **62** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **21.8** MBH
Coil CFM at Design Heating **1215** CFM
Max coil CFM **1215** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **17.0**
Ent. DB / Lvg DB **57.7 / 76.1** F

Supply Fan Sizing Data

Design CFM **1215** CFM
Design CFM/sqft **0.95** CFM/sqft

Fan motor BHP **0.67** BHP
Fan motor kW **0.53** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **335** CFM
CFM/sqft **0.26** CFM/sqft

CFM/person **26.20** CFM/person

Air System Sizing Summary for HP-102-OFFICES

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-102-OFFICES**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **774.0** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **0.7** Tons
Total coil load **8.8** MBH
Sensible coil load **8.8** MBH
Coil CFM at peak load **333** CFM
Sum of peak zone CFM **333** CFM
Sensible heat ratio **1.000**
CFM/Ton **453.1**
sqft/Ton **1054.7**
BTU/(hr sqft) **11.4**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **80.2 / 61.7** F
Leaving DB / WB **53.4 / 51.9** F
Resulting RH **50** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **6.4** MBH
Coil CFM at Design Heating **333** CFM
Max coil CFM **333** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **8.3**
Ent. DB / Lvg DB **60.7 / 80.4** F

Supply Fan Sizing Data

Design CFM **333** CFM
Design CFM/sqft **0.43** CFM/sqft

Fan motor BHP **0.18** BHP
Fan motor kW **0.14** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **73** CFM
CFM/sqft **0.09** CFM/sqft

CFM/person **13.83** CFM/person

Air System Sizing Summary for HP-103-BOH EMPLOYEE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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Air System Information

Air System Name **HP-103-BOH EMPLOYEE**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **593.7** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **1.1** Tons
Total coil load **13.3** MBH
Sensible coil load **13.3** MBH
Coil CFM at peak load **584** CFM
Sum of peak zone CFM **584** CFM
Sensible heat ratio **1.000**
CFM/Ton **526.7**
sqft/Ton **535.2**
BTU/(hr sqft) **22.4**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **September 15:00**
OA DB / WB **101.0 / 63.2** F
Entering DB / WB **76.8 / 59.3** F
Leaving DB / WB **53.7 / 50.6** F
Resulting RH **44** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **7.3** MBH
Coil CFM at Design Heating **584** CFM
Max coil CFM **584** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **12.3**
Ent. DB / Lvg DB **64.9 / 77.7** F

Supply Fan Sizing Data

Design CFM **584** CFM
Design CFM/sqft **0.98** CFM/sqft

Fan motor BHP **0.32** BHP
Fan motor kW **0.25** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **80** CFM
CFM/sqft **0.14** CFM/sqft

CFM/person **8.27** CFM/person

Air System Sizing Summary for HP-104-BOH LAUNDRY

(In Alternative: Default Alternative)

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Air System Information

Air System Name **HP-104-BOH LAUNDRY**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **1317.5** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **2.6** Tons
Total coil load **31.4** MBH
Sensible coil load **30.6** MBH
Coil CFM at peak load **1293** CFM
Sum of peak zone CFM **1293** CFM
Sensible heat ratio **0.972**
CFM/Ton **493.5**
sqft/Ton **502.9**
BTU/(hr sqft) **23.9**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **August 15:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **77.8 / 62.5** F
Leaving DB / WB **53.9 / 53.8** F
Resulting RH **55** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **14.3** MBH
Coil CFM at Design Heating **1293** CFM
Max coil CFM **1293** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **10.9**
Ent. DB / Lvg DB **64.5 / 75.8** F

Supply Fan Sizing Data

Design CFM **1293** CFM
Design CFM/sqft **0.98** CFM/sqft

Fan motor BHP **0.71** BHP
Fan motor kW **0.56** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **188** CFM
CFM/sqft **0.14** CFM/sqft

CFM/person **18.99** CFM/person

Air System Sizing Summary for HP-105-LOBBY/MARKET

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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Air System Information

Air System Name **HP-105-LOBBY/MARKET**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **1301.9** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **2.5** Tons
Total coil load **29.6** MBH
Sensible coil load **29.0** MBH
Coil CFM at peak load **918** CFM
Sum of peak zone CFM **918** CFM
Sensible heat ratio **0.980**
CFM/Ton **372.0**
sqft/Ton **527.7**
BTU/(hr sqft) **22.7**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **83.6 / 63.3** F
Leaving DB / WB **51.7 / 51.6** F
Resulting RH **55** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **23.4** MBH
Coil CFM at Design Heating **918** CFM
Max coil CFM **918** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **18.0**
Ent. DB / Lvg DB **55.2 / 81.3** F

Supply Fan Sizing Data

Design CFM **918** CFM
Design CFM/sqft **0.70** CFM/sqft

Fan motor BHP **0.50** BHP
Fan motor kW **0.40** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **298** CFM
CFM/sqft **0.23** CFM/sqft

CFM/person **10.17** CFM/person

Air System Sizing Summary for HP-106-DINING LARGE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-106-DINING LARGE**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **1092.2** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **2.9** Tons
Total coil load **34.6** MBH
Sensible coil load **34.6** MBH
Coil CFM at peak load **833** CFM
Sum of peak zone CFM **833** CFM
Sensible heat ratio **1.000**
CFM/Ton **289.0**
sqft/Ton **379.2**
BTU/(hr sqft) **31.6**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **93.7 / 63.6** F
Leaving DB / WB **51.6 / 48.0** F
Resulting RH **44** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **34.5** MBH
Coil CFM at Design Heating **833** CFM
Max coil CFM **833** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **31.6**
Ent. DB / Lvg DB **39.4 / 81.9** F

Supply Fan Sizing Data

Design CFM **833** CFM
Design CFM/sqft **0.76** CFM/sqft

Fan motor BHP **0.46** BHP
Fan motor kW **0.36** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **524** CFM
CFM/sqft **0.48** CFM/sqft

CFM/person **12.00** CFM/person

Air System Sizing Summary for HP-107-BUFFET/SERVING

(In Alternative: Default Alternative)

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Air System Information

Air System Name **HP-107-BUFFET/SERVING**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **747.0** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **1.9** Tons
Total coil load **23.3** MBH
Sensible coil load **23.3** MBH
Coil CFM at peak load **780** CFM
Sum of peak zone CFM **780** CFM
Sensible heat ratio **1.000**
CFM/Ton **401.0**
sqft/Ton **384.3**
BTU/(hr sqft) **31.2**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **82.4 / 61.5** F
Leaving DB / WB **52.1 / 50.3** F
Resulting RH **47** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **14.9** MBH
Coil CFM at Design Heating **780** CFM
Max coil CFM **780** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **20.0**
Ent. DB / Lvg DB **57.1 / 76.7** F

Supply Fan Sizing Data

Design CFM **780** CFM
Design CFM/sqft **1.04** CFM/sqft

Fan motor BHP **0.43** BHP
Fan motor kW **0.34** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **224** CFM
CFM/sqft **0.30** CFM/sqft

CFM/person **22.50** CFM/person

Air System Sizing Summary for HP-108-FOOD PREP

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-108-FOOD PREP**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **695.6** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **3.0** Tons
Total coil load **36.6** MBH
Sensible coil load **36.6** MBH
Coil CFM at peak load **1080** CFM
Sum of peak zone CFM **1080** CFM
Sensible heat ratio **1.000**
CFM/Ton **354.5**
sqft/Ton **228.3**
BTU/(hr sqft) **52.6**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **87.8 / 61.9** F
Leaving DB / WB **53.5 / 49.1** F
Resulting RH **42** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **26.6** MBH
Coil CFM at Design Heating **1080** CFM
Max coil CFM **1080** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **38.2**
Ent. DB / Lvg DB **48.6 / 73.9** F

Supply Fan Sizing Data

Design CFM **1080** CFM
Design CFM/sqft **1.55** CFM/sqft

Fan motor BHP **0.59** BHP
Fan motor kW **0.47** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **487** CFM
CFM/sqft **0.70** CFM/sqft

CFM/person **35.00** CFM/person

Air System Sizing Summary for HP-109-DINING FLEX

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
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Air System Information

Air System Name **HP-109-DINING FLEX**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **362.7** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **1.1** Tons
Total coil load **12.9** MBH
Sensible coil load **12.9** MBH
Coil CFM at peak load **361** CFM
Sum of peak zone CFM **361** CFM
Sensible heat ratio **1.000**
CFM/Ton **337.4**
sqft/Ton **338.6**
BTU/(hr sqft) **35.4**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **August 15:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **88.9 / 62.5** F
Leaving DB / WB **52.8 / 49.2** F
Resulting RH **44** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **11.4** MBH
Coil CFM at Design Heating **361** CFM
Max coil CFM **361** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **31.5**
Ent. DB / Lvg DB **47.0 / 79.4** F

Supply Fan Sizing Data

Design CFM **361** CFM
Design CFM/sqft **1.00** CFM/sqft

Fan motor BHP **0.20** BHP
Fan motor kW **0.16** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **174** CFM
CFM/sqft **0.48** CFM/sqft

CFM/person **12.00** CFM/person

Air System Sizing Summary for HP-110-DINING SMALL

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-110-DINING SMALL**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **713.0** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **2.0** Tons
Total coil load **24.5** MBH
Sensible coil load **24.5** MBH
Coil CFM at peak load **703** CFM
Sum of peak zone CFM **703** CFM
Sensible heat ratio **1.000**
CFM/Ton **344.1**
sqft/Ton **349.1**
BTU/(hr sqft) **34.4**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **August 14:00**
OA DB / WB **104.9 / 65.8** F
Entering DB / WB **88.5 / 62.4** F
Leaving DB / WB **53.2 / 49.3** F
Resulting RH **44** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **22.4** MBH
Coil CFM at Design Heating **703** CFM
Max coil CFM **703** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **31.4**
Ent. DB / Lvg DB **46.8 / 79.5** F

Supply Fan Sizing Data

Design CFM **703** CFM
Design CFM/sqft **0.99** CFM/sqft

Fan motor BHP **0.38** BHP
Fan motor kW **0.31** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **342** CFM
CFM/sqft **0.48** CFM/sqft

CFM/person **12.00** CFM/person

Air System Sizing Summary for HP-111-MEETING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-111-MEETING**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **808.8** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **2.4** Tons
Total coil load **29.4** MBH
Sensible coil load **28.4** MBH
Coil CFM at peak load **1281** CFM
Sum of peak zone CFM **1281** CFM
Sensible heat ratio **0.968**
CFM/Ton **523.6**
sqft/Ton **330.6**
BTU/(hr sqft) **36.3**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **September 14:00**
OA DB / WB **99.7 / 62.8** F
Entering DB / WB **76.7 / 62.3** F
Leaving DB / WB **54.3 / 54.1** F
Resulting RH **56** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **15.3** MBH
Coil CFM at Design Heating **1281** CFM
Max coil CFM **1281** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **18.9**
Ent. DB / Lvg DB **64.7 / 77.0** F

Supply Fan Sizing Data

Design CFM **1281** CFM
Design CFM/sqft **1.58** CFM/sqft

Fan motor BHP **0.70** BHP
Fan motor kW **0.56** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **180** CFM
CFM/sqft **0.22** CFM/sqft

CFM/person **6.84** CFM/person

Air System Sizing Summary for HP-112-ENGINEERING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-112-ENGINEERING**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **232.6** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **0.2** Tons
Total coil load **2.4** MBH
Sensible coil load **2.4** MBH
Coil CFM at peak load **77** CFM
Sum of peak zone CFM **77** CFM
Sensible heat ratio **1.000**
CFM/Ton **386.0**
sqft/Ton **1159.9**
BTU/(hr sqft) **10.3**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 16:00**
OA DB / WB **106.2 / 66.2** F
Entering DB / WB **82.9 / 62.5** F
Leaving DB / WB **51.4 / 51.1** F
Resulting RH **52** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **1.5** MBH
Coil CFM at Design Heating **77** CFM
Max coil CFM **77** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **6.5**
Ent. DB / Lvg DB **56.4 / 76.6** F

Supply Fan Sizing Data

Design CFM **77** CFM
Design CFM/sqft **0.33** CFM/sqft

Fan motor BHP **0.04** BHP
Fan motor kW **0.03** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **23** CFM
CFM/sqft **0.10** CFM/sqft

CFM/person **12.50** CFM/person

Air System Sizing Summary for HP-STAIRS1

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

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Air System Information

Air System Name **HP-STAIRS1**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **992.6** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **1.0** Tons
Total coil load **11.7** MBH
Sensible coil load **11.7** MBH
Coil CFM at peak load **512** CFM
Sum of peak zone CFM **512** CFM
Sensible heat ratio **1.000**
CFM/Ton **523.8**
sqft/Ton **1016.2**
BTU/(hr sqft) **11.8**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 13:00**
OA DB / WB **102.9 / 65.2** F
Entering DB / WB **73.0 / 55.2** F
Leaving DB / WB **49.7 / 45.5** F
Resulting RH **33** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **8.4** MBH
Coil CFM at Design Heating **512** CFM
Max coil CFM **512** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **8.5**
Ent. DB / Lvg DB **72.0 / 88.9** F

Supply Fan Sizing Data

Design CFM **512** CFM
Design CFM/sqft **0.52** CFM/sqft

Fan motor BHP **0.28** BHP
Fan motor kW **0.22** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **0** CFM
CFM/sqft **0.00** CFM/sqft

CFM/person **0.00** CFM/person

Air System Sizing Summary for HP-STAIRS2

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)
Prepared by: Shakespeare Engineering

05/29/2026
1:21 PM

Air System Information

Air System Name **HP-STAIRS2**
Equipment Class **SPLT AHU**
Air System Type **SZCAV**

Number of zones **1**
Floor Area **1000.7** sqft
Location **St George Municipal, UT, USA**

Sizing Calculation Information

Calculation Months **Jan to Dec**
Sizing Data **Calculated**

Zone CFM Sizing **Sum of space airflow rates**
Space CFM Sizing **Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load **1.2** Tons
Total coil load **14.3** MBH
Sensible coil load **14.3** MBH
Coil CFM at peak load **607** CFM
Sum of peak zone CFM **607** CFM
Sensible heat ratio **1.000**
CFM/Ton **509.7**
sqft/Ton **840.5**
BTU/(hr sqft) **14.3**
Water flow @ 10.0 F rise **N/A**

Peak coil load occurs at **July 19:00**
OA DB / WB **100.1 / 64.3** F
Entering DB / WB **73.0 / 55.2** F
Leaving DB / WB **49.1 / 45.2** F
Resulting RH **33** %
Design supply temp. **55.0** F
Zone T-stat Check **1 of 1** OK
Max zone temperature deviation **0.0** F

Central Heating Coil Sizing Data

Max coil load **8.6** MBH
Coil CFM at Design Heating **607** CFM
Max coil CFM **607** CFM
Water flow @ 20.0 F drop **N/A**

Load occurs at **Design Heating**
BTU/(hr sqft) **8.5**
Ent. DB / Lvg DB **72.0 / 86.5** F

Supply Fan Sizing Data

Design CFM **607** CFM
Design CFM/sqft **0.61** CFM/sqft

Fan motor BHP **0.33** BHP
Fan motor kW **0.26** kW
Fan total static **2.00** in wg

Outdoor Ventilation Air Data

Design airflow CFM **0** CFM
CFM/sqft **0.00** CFM/sqft

CFM/person **0.00** CFM/person